

Jetson LLM Figure-Matching Experiment - Short Notes

Compact summary of current observations and client-side prompt design.

1. Main issue observed

- Jetson agents hallucinated and looped more when the prompt gave prescribed responses and detailed step-by-step instructions.
- The model often copied a response pattern instead of reasoning about the current game state.
- For gemma3:4b, more instructions did not mean better behavior; the prompt became too heavy.

2. Why long history was a problem

- Too much conversation history made the agents focus on old messages and repeat earlier patterns.
- Old proposals or candidate lists could confuse the model about the current stage.
- Current version gives only the last 7 public messages instead of the full history.

3. What each Jetson sees before responding

- Its assigned agent name, total number of agents, and private figures.
- The current proposed figure, if there is one.
- Which agents already confirmed that proposed figure.
- Whether this agent already listed its figures.
- The last 7 public messages from the current trial.

4. Current prompt design

- The prompt is shorter and less scripted than earlier versions.
- It still gives the goal and basic rules: do not invent figures, use only private figures and public messages, and submit only with enough support.
- It gives possible actions, but lets the model decide what is useful to say next.

5. Current prompt text

- Attached below is the current system prompt used in new_client.py.

```
You are {agent_name}, one of {num_agents} agents in a figure-matching experiment.

YOUR FIGURES:
{symbols_str}

TASK:
There is exactly ONE figure that is shared by ALL {num_agents} agents.
Your goal is to help the group identify that common figure.

RULES:
- You can see only YOUR figures and the public messages.
- Do not invent figures.
- Do not claim you have a figure unless it is in YOUR FIGURES.
- Do not submit a final answer unless the public conversation gives enough support.
- Keep your message short and relevant.
```

STRATEGY:

- Start by telling the other agents what symbols you have.
- When other agents share their symbols, compare their symbols with yours.
- The common symbol is the one that appears in EVERY agent's list.
- If you think you found the common figure, propose it using this exact phrase: I propose <figure>.
- Once you identify it and believe there is enough support, submit the full word.

IMPORTANT:

- Always use the exact format below: ACTION: CHAT or ACTION: ANSWER.
- Keep messages short and direct.
- Only submit ANSWER when you are sure.

You may decide what is useful to say next.

6. Additional state extracted by the client

- `get_current_proposal()` finds the latest message saying "I propose ".
- `get_confirmed_agents()` finds agents who said "I confirm ".
- `agent_already_listed()` checks whether this same agent already said "My figures are:".

7. Current model settings

- Temperature = 0.0: makes output more deterministic, but does not guarantee correct reasoning.
- Top-p = 0.8: limits unusual word choices.
- Repeat penalty = 1.2: tries to reduce repeated loops.
- Num predict = 80: keeps responses short.

8. Important experiment point

- The code still lets the LLM decide what to say.
- The client parses ACTION: CHAT or ACTION: ANSWER, but it does not hardcode the correct answer.
- This keeps the experiment meaningful because hallucinations and coordination failures remain visible.

9. Current conclusion

- The newer setup is better because it gives the model less history, clearer state, and more freedom.
- This is a better test of actual LLM coordination than using hardcoded correction rules.
- The tradeoff is that responses may be less predictable, but the observed behavior is more honest for the experiment.

Source: current `new_client.py` uploaded in this conversation.